



# How Blockchain Can Be Effectively Leveraged By The Telecommunication Industry

# Abstract:

Although blockchain as a technology has been present for a decade, it has only gained traction when industries such as Banking and Financial Services, retail, and logistics successfully implemented different blockchain use cases to address their pain points. The technology's success in these industries has catalyzed investment in other domains such as telecommunications. Blockchain can unlock financial and operational synergies by providing efficiencies in transactions between Communication Service Providers and third parties such as interconnected partners. The technology can significantly impact the telecommunication industry. The benefits of deploying blockchain-based solutions will improve business operations. In particular, the benefits will accrue in the areas of network & contract management, payments and settlements, and identity management among others. Keeping in mind the technology's transformational abilities, telecommunication players are well advised to be among the early movers. A suggested way forward is to partner with competent companies capable of suggesting innovative use cases for their quick trial and implementation.

## **Author:**

Amol Shrikant Kulkarni - Lead Business Analyst,  
Sasken Technologies Limited.



# Table of Content

|                                                                                |    |
|--------------------------------------------------------------------------------|----|
| An Introduction to Blockchain .....                                            | 04 |
| Impact on Telecommunication Industry and Communication Service Providers ..... | 06 |
| Use Cases for Blockchain in Telecommunication .....                            | 08 |
| Current Blockchain Consortia .....                                             | 12 |
| Key Implications for Telecommunication Executives .....                        | 13 |
| Conclusion .....                                                               | 14 |
| About The Author .....                                                         | 14 |
| About Sasken .....                                                             | 14 |



# An Introduction to Blockchain

Blockchain is a distributed ledger, wherein all transactional records are cryptographically signed and immutable. Each record is time-stamped and has linkages to the previous transactions. The records are grouped into blocks and added to the chain of prior blocks successively created. The transaction records are shared with all the participating entities, and each participant in the network can ensure the veracity of transactions based on a consensus protocol. Network consensus methods and cryptographic techniques are used to irrefutably validate transactions.

The main capabilities of blockchain network include the ability to:

- Represent multiple types of assets digitally
- Enable new forms of value exchanges
- Transact with no central authority or middleman
- Ensure distributed, immutable, traceable copies of identical records and assets
- Enhances trust within the ecosystem due to its immutable nature

These attributes enable governance, management, and execution of assets across entities, dynamically and autonomously. Blockchain as a technology was first introduced to the industry as means to manage and protect bitcoin and other cryptocurrencies. However, within a few years, it gained traction and wide adoption across industries. The adoption of technology is expected to grow exponentially. [Gartner expects that the technology will generate \\$176 billion \(by 2025\) and more than \\$3.1 trillion \(by 2030\) of benefits and value to businesses across all industries.](#)

The blockchain ecosystem currently has the following parties:

- Upstream players: Platform developers such as Corda, Hyperledger, and Quorum along with consulting & service providers such as KPMG, PWC, Accenture, Infosys, TCS, etc.,

- Application Types: Blockchain has multiple applications; for example, Smart Contracts can be integrated into legal agreements, payment transfer regulations, and IoT gateway management. It can also help with identity management for self-sovereign and digital identity applications
- Downstream Players: Consist of industries implementing the use cases - Banking, Financial Services, and Insurance (BFSI) organizations, telecommunication operators, retail organizations, government bodies, healthcare specialists, etc.

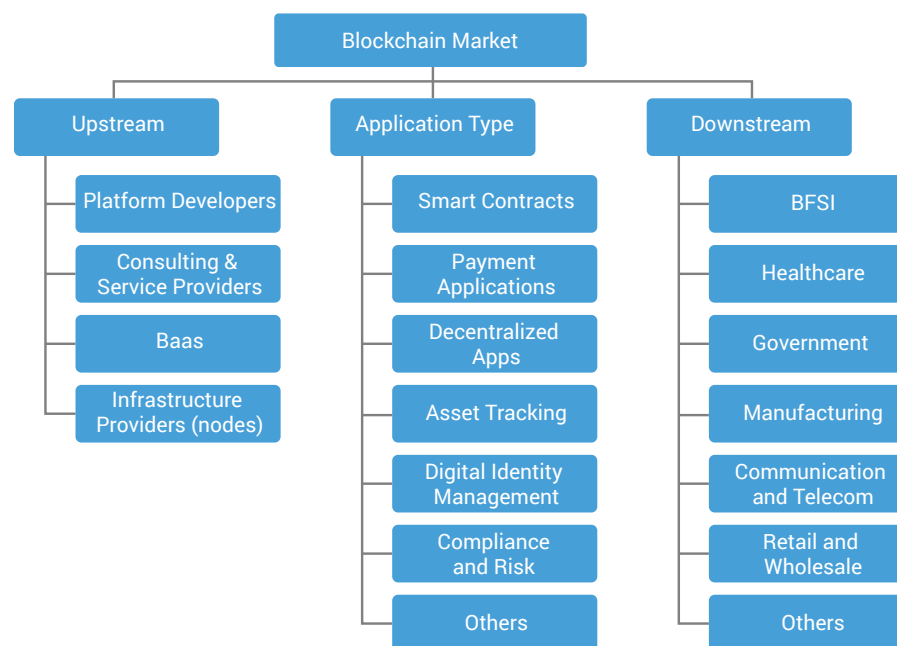


Figure 1: The Present Blockchain Ecosystem



The downstream players align with upstream entities to develop various applications such as smart contracts and decentralized applications. These form part of different use cases, including asset tracking, provenance checks, payments, and settlements.

Provided below is a list of use cases being pursued by different industries:

| Industry                   | Asset Tracking             | Identity Management             | Shared Record Keeping               | Loyalty and Rewards      | Payment & Settlement        | Provenance              | Trading & Voting                    |
|----------------------------|----------------------------|---------------------------------|-------------------------------------|--------------------------|-----------------------------|-------------------------|-------------------------------------|
| Banking                    | -                          | KYC Management                  | Accounting and Bookkeeping          | Credit Card Spending     | Cross Border Payment        | -                       | Trade Finance                       |
| Securities                 | Securities Lending         | Customer KYC                    | -                                   | -                        | Dividend Payments           | -                       | E-Voting for bonds and stakeholders |
| Logistics & Transportation | Location intelligence      | Tag Authentication              | Port Clearance                      | -                        | Distributor Payments        | Part Authenticity       | Shipping Trade Finance              |
| Retail and Wholesale       | Food Tracking              | Counterfeit Products Prevention | Inventory Management                | Partner Spending Rewards | Crypto Payments             | Source of Origin (Farm) | -                                   |
| Telecom                    | -                          | E-Sim                           | 5G Device authentication            | -                        | Roaming Settlements         | Fraud Prevention        | -                                   |
| Automotive                 | Aftermarket parts Tracking | Vehicle Identity                | Vehicle data sharing to aggregators | -                        | Parking Payments (Mobility) | EV Batteries Source     | -                                   |
| Healthcare                 | Vaccine Tracking           | Digital Health Cards            | Electronic Health Records           | -                        | Insurance Payments          | Drug Counterfeit        | -                                   |
| Government                 | Food Supply                | Citizen Identity                | Passport Data Sharing               | -                        | Pension Settlements         | Credential Verification | E-Voting                            |
| Education                  | -                          | Student Credit Records          | Science Paper Publishing            | Credit Transfer          | Education Loan              | Transcripts Management  | -                                   |

Popular Use Cases

Figure 2: Blockchain Use Cases by Verticals



# Impact on Telecommunication Industry and Communication Service Providers

The telecommunication value chain for both consumers and businesses can be categorized into three parts: Network Infrastructure, Data and Voice Connectivity, and Consumer Services. Communication Service Providers (CSP) like AT&T, Telefonica, and Vodafone have traditionally owned the end-to-end telecom value chains. However, with recent shifts in consumer preferences, including data explosion, the transition to digital infrastructure, and heightened competition from new OTT players, CSPs are witnessing a decrease in revenues from voice applications. It is also leading to an increase in costs due to the need to provision high-volume data-oriented bandwidth networks. CSPs are therefore exploring avenues that reduce costs and generate new sources of revenue.

There are potential use cases of blockchain, both for current operations management and for future telecommunication applications. CSPs are, therefore, likely to adopt blockchain in their core operational management and its adjacent areas, as seen in the figure below:

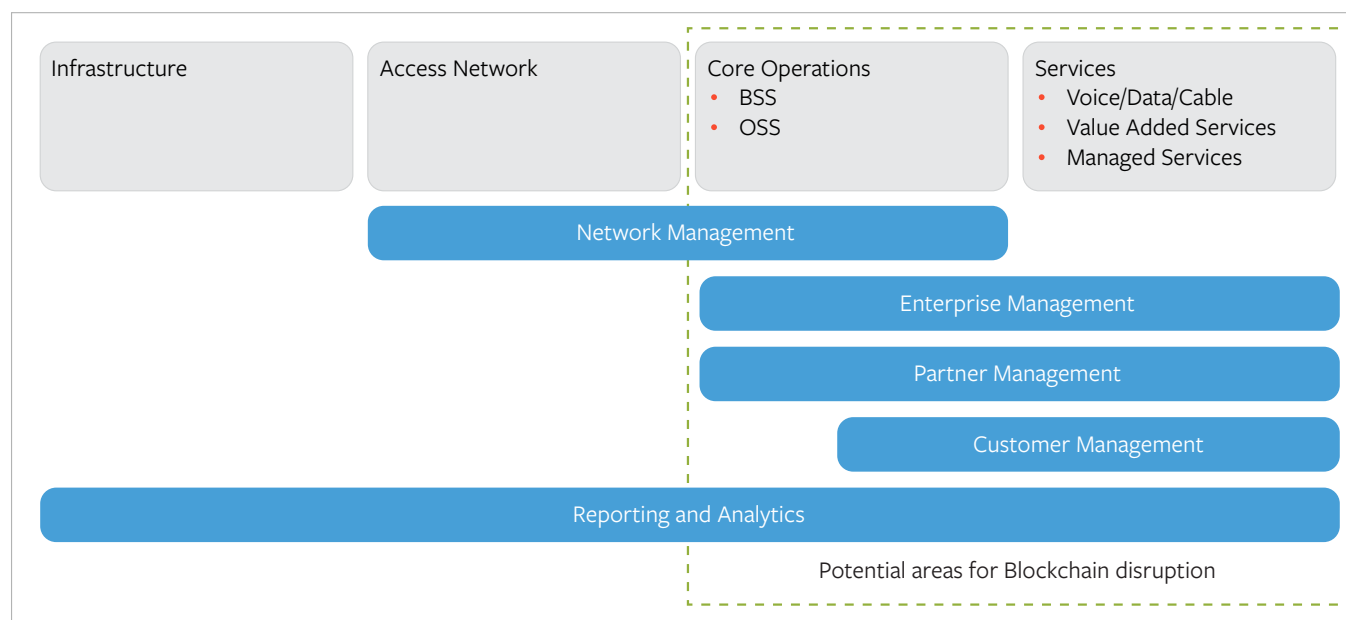


Figure 3: Blockchain impact on Communication Service Provider | Source: [Deloitte](#)



Provided in the following chart are few impactful use cases, which telecommunication operators can leverage blockchain to increase efficiency in the ecosystem:

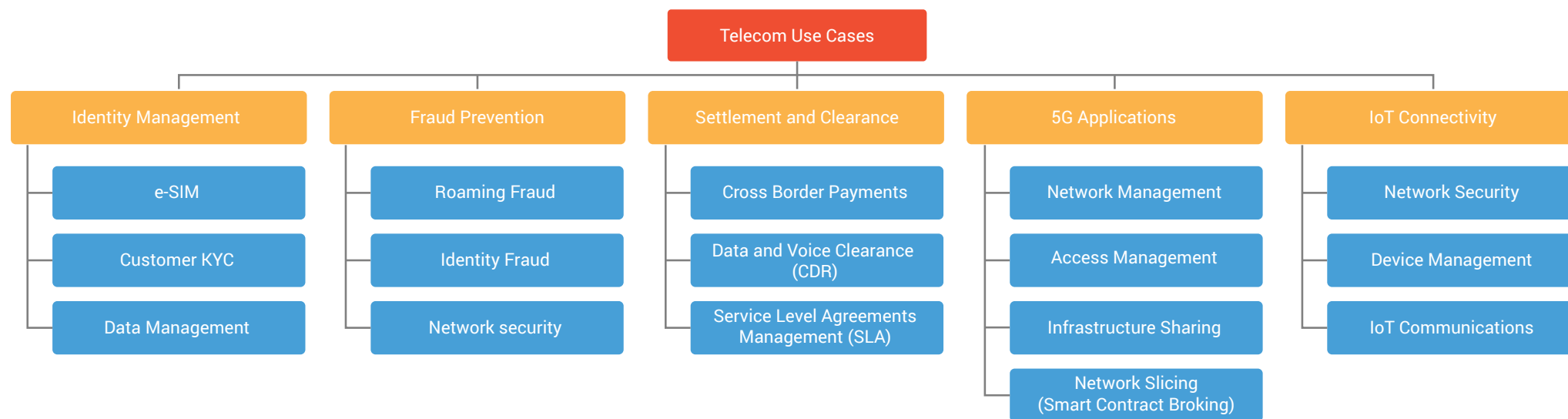


Figure 4: Use Cases for Blockchain in the Telecommunication Industry

We will be looking at Identity Management and Fraud Prevention as use cases in detail in this paper.



# Use Cases for Blockchain in Telecommunication

## a. Identity Management: e-SIM

An e-SIM is nothing but a permanently installed embedded SIM card in each device.

Telecommunication providers are moving from traditional SIM cards to e-SIM cards to leverage the advantages of digital technologies. Benefits such as compliance with GSMA standards across all operators, users' ability to avail number portability, and elimination of SIM card slots resulting in additional savings. McKinsey predicts that the market for e-SIMs will reach \$5.6 billion in 2022 as compared to the \$0.6 billion projection in 2017, with a CAGR of 95%. The identity management aspects of SIM cards can be efficiently handled using blockchain's capabilities.

### Current System

Currently, if a user wants to register at a CSP vendor, they need to provide appropriate documents to prove their identity. Each time users submit all their information to vendors, even if only specific parameters are required. End-users may also share additional information and generate multiple login credentials while signing up for services online. In this scenario, users, and telecommunication operators have to manage numerous information parameters simultaneously, leading to duplication of information at each data sharing occasion.

### Blockchain-based System

A blockchain-based solution can be utilized as a shared ledger to store descriptions of all identity parameters. CSPs can act as an identity-as-a-service provider for their partners and monetize this identity data, thereby creating an additional revenue stream.

In this system, by following an e-SIM ecosystem, when a user registers on the CSP network, the CSP creates a unique digital identity for the user. The identity is based on blockchain cryptography characteristics. This digital identity contains a private key, which is eventually stored in e-SIM. The CSP then creates a virtual identity based on the public key of a user's digital identity and encrypts it using a digital signature based on the CSP's private key. The identity is then added to CSP's blockchain network, as showcased in the following figure.

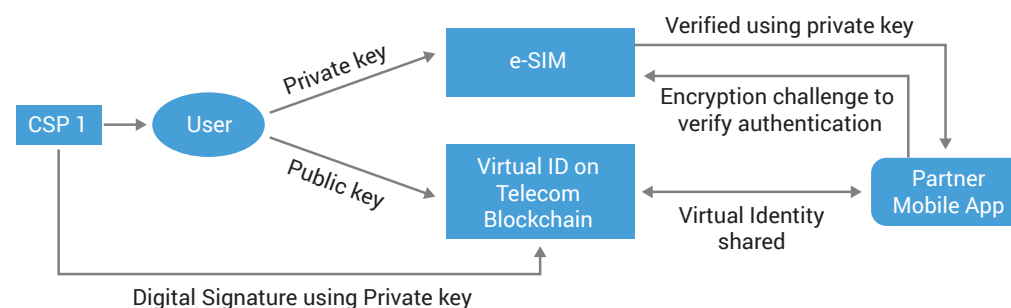


Figure 5: Blockchain-based e-SIM identity management



If the user visits the CSP's partner's website, for example, an e-commerce site, then the phone application connected to CSP's blockchain identifies the user's virtual identity. The user's identity is authenticated by verifying the private key in the e-SIM. The partner can now create its own virtual identity for the user based on these parameters. Aspects such as the user's search history and purchase history can be stored in their virtual identity, which is a part of the partner's blockchain database. The partner can also use this to provide personalized recommendations to the end-user through its analytics engine. Additionally, the user can log in to other partners' mobile applications using the same mechanism. Hence, the cost and time associated with providing valid identity proof, its authentication, and KYC can be mitigated by the CSP partner's ecosystem.

CSPs can also expand their ecosystem by providing and validating user identities for various industries such as education institutes for certificates, travel agencies for passport verification, and enterprises for identity verification.

CSPs could receive the following benefits by adopting this system:

- Cost savings by adopting a blockchain-based identity management platform
- Additional revenue via sharing of users' identity data with their partners' ecosystems
- New data sharing avenues leading to expansion of partners' ecosystems
- An enhanced subscription experience for end-users leading to an increase in customer satisfaction and subsequently in its retention rates

### **b. Fraud Management: Mitigating Roaming Fraud**

Fraud management has been one of the most crucial and challenging aspects to tackle for the telecommunication industry. Frauds related to roaming, identity theft, and payments cost the telecom industry over \$38 billion every year. The industry has not found a single effective solution to prevent fraud. The most common type of fraud occurs during roaming as users latch on to the Visited Public Mobile Network (VPMN). The fraud occurs when a user moves from his existing home network to another operator's network and then uses their services with no intention of paying, making the home network responsible for the charges. Blockchain technology, if integrated with existing networks, can significantly reduce the overall costs of roaming fraud.

#### **Current System**

When a user visits a different region or country, the mobile device communicates its IMSI number to the Visited Public Mobile Network's (VPMN) Visitor Local Register (VLR). When a user calls any number, the VPMN sends inquiries to the Host Public Mobile Network (HPMN) about the roaming services that the user is using. The user's subscription details are then queried to the Home Location Register (HLR), and details of call records are sent to the billing systems HPMN in the form of a Transfer Account Procedure (TAP) file. An intermediary company, i.e., the Data Clearing House (DCH), transfers/converts the TAP files for the CSP's HPMN. Further, the HPMN settles the accounts with the VPMN as per their roaming tariff agreements.



Roaming fraud is generally perpetrated in multiple ways, but the most common characteristics are:

- Time Delays- Interchange of roaming data between networks is a time-consuming process. Also, the time required to detect and respond to frauds is longer due to delays in data exchange
- Multiple Subscriptions- Fraudulent access to SIM cards from numerous operators is a prevailing issue
- Security Attacks- The industry's current encryption method – SS7 encryption – can be compromised using brute force attacks

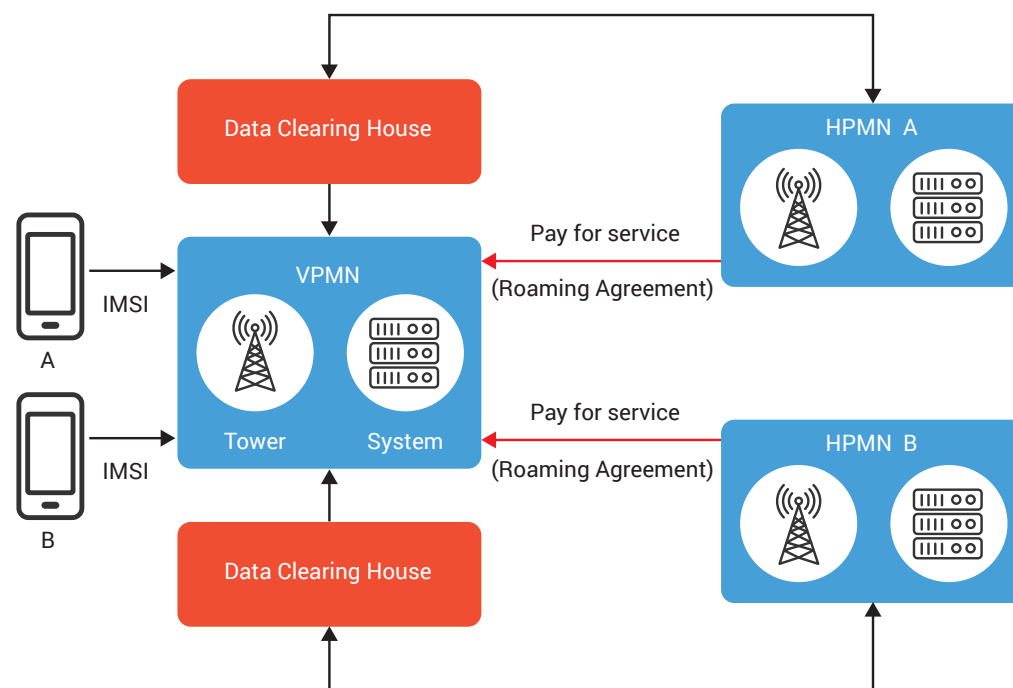


Figure 4: The current system leveraging VPMN and HPMN for roaming-based transaction



### Blockchain-based System

The current system has limited authentication mechanisms for users and does not validate data in real-time. To mitigate present risks & challenges, integrating a permission-seeking form of blockchain could help address the issue. Each network operator will have designated blockchain nodes that verify the integrity of the transactions being broadcast on the network. The roaming agreements can be embedded in smart contracts, triggered when a data transfer request is initiated.

Whenever a user calls/triggers an event through the VPMN network, it sends the call data records to HPMN through a smart contract, where the roaming agreement is executed. The HPMN node validates the transaction, and the bill is generated automatically.

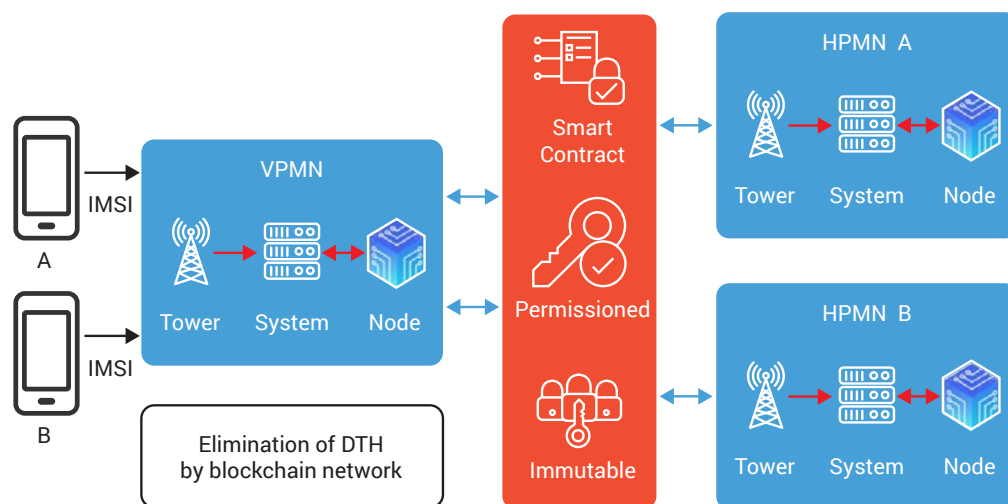


Figure 5: A proposed blockchain-based system leveraging VPMN and HPMN for roaming-based transaction

The modified blockchain-based system ensures real-time verification, authorization, and settlements. In this case, the telecommunication value chain can undergo a consolidation phase, wherein CSPs can also do away with DCHs as middlemen, resulting in additional cost savings.

The benefits of a blockchain-based system in this use case are:

- Cost savings by eliminating Data Clearing Houses
- Instantaneous bill settlement reducing the duration of cash flow cycles
- Quick resolution of the settlement disputes and hence reduction in relevant operational costs



## Current Blockchain Consortia

To reap the benefits of blockchain, the participation of all players in the value chain is critical. The consortia-based model drives the adoption of blockchain in the ecosystem. Communications Blockchain Network (CBAN) and Carrier Blockchain Study Group (CBSG) are two significant consortiums providing blockchain implementation in the telecommunication industry.

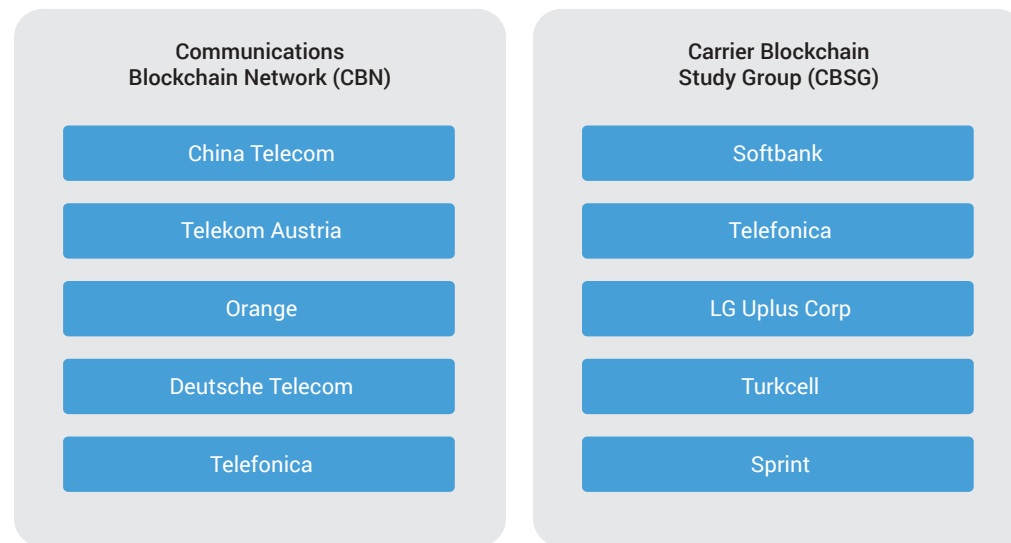


Table 3: Key Consortia in Telecom Segment

**Communications Blockchain Network (CBAN):** Founded by ITW Global Leader's Forum (GLF) – a group of telecom industry leaders came together to explore the possibility of a real-time, blockchain-enabled platform that can revolutionize the manually cumbersome and isolated practices around inter-carrier settlement processes, which so far have cost the industry billions.

**Carrier Blockchain Study Group (CBSG):** The platform was launched in September 2017 to jointly develop innovative blockchain platforms, specifically designed for telecom carriers. It is led by TBCASoft (US based blockchain technology company), and SoftBank Corp., along with Sprint Corporation (US), and FarEasTone Telecommunications Co. (Taiwan) as initial founding members. CBSG aims to develop innovative services such as optimized transaction records, identification and authentication, secure mobile transactions, IoT applications and other services.



# Key Implications for Telecommunication Executives

Blockchain as a technology can improve the top line and bottom line of organizations in the telecommunication ecosystem.

## 1. Cost Reduction and Optimization:

- Blockchain can substantially reduce the cost of operations (KYC, Identity Management) and eliminate intermediaries such as DCHs for roaming settlements
- Reduction in roaming, identity, and subscription frauds. [For example, eliminating roaming frauds impact, which is ~15% of a CSP's operating margin.](#)
- The introduction of blockchain-based e-SIM would also eliminate SIM card manufacturing costs
- Immutable data trails and smart contracts would reduce time and costs associated with dispute management, such as legal and audit fees. Especially considering that roaming dispute settlement can take up to 30 days for a resolution
- Early settlements lead to a reduction in the cash flow cycle. Thereby the cost of servicing operations can be optimized

## 2. New Revenue Streams:

- CSPs can enter a service-based business model with offerings such as Identity-as-a-Service and Network-as-a-Service
- The partner ecosystem of CSPs could also expand to include non-allied telecommunication industries such as education and travel
- Telecommunication operators could also enter the domain of financial services by leveraging Blockchain for digital wallets, as seen with the Telcoin platform

## 3. Advancements in Technology

- Security is one of the fundamental propositions of this technology. Blockchain can be used to enhance data and network security
- Blockchain's innate characteristics could also introduce innovation in 5G enablement. For example, it could facilitate fast and reliable selection of access networks and reduce latencies in the network
- For IoT devices, smart contracts can verify and authenticate thousands of devices/sensors in the network, ensuring secure and error-free transactions



## Conclusion

Even though blockchain technology is still evolving, it is predicted to reach an inflection point in FY 2023-24. The benefits provided to the telecommunication ecosystem are multifold – from cost optimization, new revenue streams, technological innovation to the consolidation of players in the value chain. CSP's can gain competitive advantage, reap the benefits of the technology by implementing the following recommendations:

- Evaluate the blockchain market, including technological advancements, platforms, service providers, regulations, and standards
- Engage with a blockchain provider – startup, services, or platform players – to understand business implications, technological requirements, and relevant use cases
- Ideate and invest in proof of concepts and evaluate its impact on revenue maximization, cost reduction, and improving internal efficiencies

## About the Author

Amol has a decade of experience in the Manufacturing and Automotive industries. At Sasken, he works for the Product Engineering Practice and is responsible for developing and marketing new services and offerings for multiple domains.

## About Sasken

Sasken is a specialist in Product Engineering and Digital Transformation providing concept-to-market, chip-to-cognition R&D services to global leaders in Semiconductor, Automotive, Industrials, Consumer Electronics, Enterprise Devices, SatCom, Telecom, and Transportation industries. For over 30 years and with multiple patents, Sasken has transformed the businesses of 100+ Fortune 500 companies, powering more than a billion devices through its services and IP.





# How Blockchain Can Be Effectively Leveraged By The Telecommunication Industry

[marketing@sasken.com](mailto:marketing@sasken.com) | [www.sasken.com](http://www.sasken.com)

USA | UK | FINLAND | GERMANY | JAPAN | INDIA

© **Sasken Technologies Ltd.** All rights reserved.

Products and services mentioned herein are trademarks and service marks of **Sasken Technologies Ltd.**, or the respective companies.